

Task Description for a Bachelor Thesis on

Open Source Linear-Exponential-Algorithm Based on a Custom Dive Computer

at the Department of Information Technology and
Electrical Engineering

for

Sebastian Thomas
tsebastian@student.ethz.ch

Advisors: Tommaso Polonelli,
tommaso.polonelli@pbl.ee.ethz.ch

Professor: Prof. Dr. Luca Benini, lbenini@iis.ee.ethz.ch

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1 Project Goals

Dive Computers are used in SCUBA diving primarily to measure depth and time of the current immersion and compute tissue loading or bubble growth to allow divers to safely execute multi-level dives beyond no-decompression limits (NDL) by reducing the risk of decompression sickness (DCS) as well as pulmonary and CNS oxygen toxicity. Most modern Dive Computers are based on the tissue models (Bühlmann ZHL-16 (B/C)) or Bubble Models (e.g. the Reduced Gradient Bubble Model, RGBM), while use of the latter is less frequent. They primarily consist of one or more depth sensors, accurate real-time clocks, a computing unit and a display.

The Thalmann decompression algorithm developed by decompression scientists in the US Navy has been designed for deep diving beyond 100m based on an iso-risk model but is not available in any commercial dive computers. Next to the specific tables of parameters published, the main idea of the algorithm is to limit the modeled amount of decompression in all tissues while supersaturation is present in the tissues by employing a linear function with high and an exponential for lower supersaturation, while the Bühlmann algorithm only uses an exponential function even for decompressing in high supersaturation. The iso-risk Thalmann algorithm makes deep diving DCS incidence more predictable, keeping the risk constant independent of maximum dive depth and thus maximum supersaturation.

Virtually all commercial dive computers for SCUBA divers are battery-powered embedded systems, which makes energy management crucial to allow divers to make multiple and longer dives on a single charge, while keeping the form factor practicable. In broader embedded systems research, adaptive sample rates for components and calculation rates for algorithms have been researched and shown to significantly reduce energy consumption by increasing idle phases. In SCUBA diving, most dives consist of ascent- and descent stages as well as constant-depth stages (only small variations in depth). These stages have different demands on the sample as well as refresh rates of components to accurately report depth and recalculate decompression obligation and oxygen toxicity limits, and no research has been published as to how this may be exploited in dive computers.

Most commercially available dive computers are developed closed source and only provide information on dive computer functions and limitations. This project attempts to design and prototype a dive computer which can be adapted to specific divers need either by configuring available software or redesigning parts of the system without having to build a complete end-to-end system of hardware and

software. By implementing multiple decompression algorithms, including the Thalmann algorithm, this prototype will be comparable to existing commercial models while allowing divers to choose to increase their safety for deeper dives. Lastly, it demonstrates adaptive sampling and refresh rates for sensors, computations and the user interface, optimizing energy efficiency without compromising safety and maintaining the required accuracy.

2 Tasks

The project will be split into four phases, as described below:

Phase 1 (Week 1-4)

1. Wearable sensor component definition. This step also include the definition of system requirements
2. Final schematic which is used as a base for firmware and algorithmic development and improvement.
3. Investigate previous works and systems which use adaptive sampling to improve energy efficiency of electrical designs.
4. Literature review and analysis of DIVE decompression algorithms.

Phase 2 (weeks 5-8)

1. Schematic and project review with the supervisor.
2. Implement firmware for the dive computer, including the selected decompression algorithm on an evaluation kit that mimic the final MCU.
3. PCB layout design
4. Implement variable sample and refresh rate algorithm, tuneable by parameters, and use its output to schedule tasks.

Phase 3 (week 9-11)

1. PCB assembly and electrical verification.
2. Test the firmware on the developed DIVE computer design.
3. Field verification of the hardware/firmware co-design in comparison with the intended project specifications.

4. Energy efficiency calculation for a set of representative dive profiles.

Phase 4 (week 12-14)

1. Result analysis.
2. Report and presentation.

Milestones

The following milestones need to be reached during the thesis:

- PCB Design and manufacturing
- Final report and presentation

3 Project Organization

During the thesis, students will gain experience in the independent solution of a technical-scientific problem by applying the acquired specialist and social skills. The grade is based on the following: student effort; thoroughness and learning curve; achieving qualitative and quantitative results with a scientific approach; supporting practical findings with theoretical background and literature investigations; final presentation and report; documentation and reproducibility. All theses include an oral presentation, a written report and are graded. The report and presentation need to have publication grade quality to achieve a good grade. Students are graded based on the official ITET grading form¹. For students of IIS (Prof. Benini) a special grading scheme exists, please contact your supervisor for details there. Before starting, the project must be registered in myStudies and all required documents need to be handed in for archiving by PBL.

3.1 Laboratory Rules

The students agree to follow the lab rules set by PBL staff, for detail please contact us. The most important points are:

- All ETH safety regulations need to be followed², in addition to ones given by PBL staff

¹https://ethz.ch/content/dam/ethz/special-interest/itet/departement/Studies/Forms/ITET_Grading_Form_2025.xlsx

²<https://ethz.ch/staffnet/en/service/safety-security-health-environment/sicherheit-in-laboren-und-werkstaetten/laborsicherheit.html>

- No device in the lab is used without introduction by your supervisor or PBL staff
- No device leaves the lab without being officially borrowed, this is done by PBL staff and needs your Legi.
- Any damage to devices or tools needs to be reported immediately to PBL staff.
- The Lab-desk is clean and free for others after you finished your task, or when you take longer breaks. All tools are correctly sorted into their drawers/cupboards when you leave.

3.2 Weekly Report

There will be a weekly report/meeting held between the student and the assistants. The exact time and location of these meetings will be determined within the first week of the project in order to fit the students and the assistants schedule. These meetings will be used to evaluate the status and document the progress of the project (required to be done by the student). Beside these regular meetings, additional meetings can be organized to address urgent issues as well. The weekly report, along with all other relevant documents (source code, datasheets, papers, etc), should be uploaded to a clouding service, such as Polybox and shared with the assistants.

3.3 Project Plan

Within the first month of the project, you will be asked to prepare a project plan. This plan should identify the tasks to be performed during the project and sets deadlines for those tasks. The prepared plan will be a topic of discussion of the first week's meeting between you and your advisers. Note that the project plan should be updated constantly depending on the project's status.

3.4 Final Report and paper

PDF copies of the final report written in English are to be turned in. Basic references will be provided by the supervisors by mail and at the meetings during the whole project, but the students are expected to add a considerable amount of their own literature research to the project ("state of the art").

3.5 Final Presentation

There will be a presentation (15 min presentation and 5 min Q&A for BT/ST and 20 min presentation and 10 min Q&A for MT) at the end of this project in order to

present your results to a wider audience. The exact date will be determined towards the end of the work.

References:

Will be provided by the supervisors by mail and at the meetings during the whole project.

Place and Date Zürich, 17.02.2025 Signature Student *S. Thomas*